

Ana Lorena Ortiz Loyola

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Data science student (Tecnológico de Monterrey, expected 2027) with direct, hands-on experience in banking-supervisory data and regulatory reporting at Deutsche Bundesbank (AnaCredit), where I built a parameterized validation framework and statistical benchmarking logic for supervisory reporting data in Python, SQL, and R. Comfortable working with Meldewesen datasets end-to-end: validation logic, anomaly detection, and auditable documentation for reporting institutions.

Core Competencies

- **Programming:** Python, R, SQL, Java, C++
- **Regulatory Reporting Data:** Parameterized validation frameworks for supervisory (Meldewesen) reporting data; percentile-based statistical benchmarking to flag anomalous reporting frequencies
- **Data Quality & Automation:** Automated migration and error-detection testing (Pandas), YAML-driven batch validation, automated ticketing and stakeholder communication
- **Databases & Tools:** SQL, SQLite, MySQL, Excel, Git, Jira
- **Machine Learning:** Scikit-Learn, Random Forest, SVM, K-Means, predictive modeling
- **Analysis & Communication:** Structured reporting for regulatory stakeholders, German (C1) and English (C1)

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**
Data Engineer / Intern *04/2026–07/2026*
 - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames to ensure identical business logic after data migration
 - Implemented automated error detection identifying deviations (duplicate IDs, value-range violations), with automated Jira ticketing and emails to reporting banks
 - Worked within an agile Scrum framework (sprint planning, daily stand-ups, sprint reviews)

Automated Validation Framework for Regulatory Reporting Data – converted a non-scalable, case-by-case validation script into a parameterized, reusable Python solution

 - Designed parameterized MySQL queries applicable to any database, attribute, and reporting period, replacing hard-coded per-case script logic
 - Extended the validation logic with percentile-based statistical benchmarking over a rolling 12-month window to flag anomalous reporting frequencies and separate false positives from genuine data-quality issues
 - Built a YAML-driven configuration interface enabling batch validation across multiple databases, attributes, and reporting periods in a single call

Test Data Generator for the Bundesbank Platform – configuration-driven system producing complete, consistent synthetic test datasets

 - Generated Cartesian-product attribute combinations per table with consistent IDs across tables to preserve referential integrity
 - Exported datasets as XML with full schema metadata and plausibility metrics

Projects

- **Personal Project (GitHub)**
○ *Banking Data Pipeline – Data Cleaning & Audit Log* 05/2026
 - Built an end-to-end pipeline for simulated bank transactions (150 accounts) with a normalized SQLite database and foreign-key integrity checks
 - Staged data cleaning with a fully auditable log documenting every change (table, column, old/new value, timestamp)
- **Tecnológico de Monterrey**
○ *Lattice-Based Cryptography Research* 10/2024–02/2025
 - Implemented and analyzed the complexity of lattice-based cryptographic algorithms (LWE, NTRU) in Python; contributed to a scientific publication
- **Tecnológico de Monterrey**
○ *Predictive Modeling – Healthcare Optimization* 10/2024–12/2024
 - Analyzed 25+ years of Mexican national health data to identify patient risk profiles and resource bottlenecks; built predictive models (Random Forest, Neural Networks) and K-Means clustering for high-risk segmentation

Education

- **Tecnológico de Monterrey** Mexico
○ *BSc in Data Science* 2022–2027
Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: data analysis & statistics, machine learning & AI, cryptography & security, numerical optimization.
- **Universität Göttingen** Germany
○ *Exchange Semester* 2025–2026

Languages

- German (C1), English (C1, TOEFL iBT 105/120), Spanish (native).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

References

- Available upon request.